single asset trading case. Moreover, in portfolio trading with mixed price trends, our within-episode risk control mechanism performs robustly by monitoring the entire portfolio's value fluctuations. The second study focuses on the over-episode risk control mechanism, demonstrating its critical role in updating the trading manager agent's beliefs to provide a more comprehensive understanding of current trading conditions, as articulated in Table 5. The markedly improved CRs and SRs in both decision-making scenarios underscore the effectiveness of using CVRF to update investment beliefs episodically, guiding the agent towards more profitable investment strategies. Additionally, FINCON demonstrates significant learning gains, achieving these results after only four training episodes—substantially fewer than what is typically required by traditional RL algorithmic trading agents. More visualizations and analysis are provided in the Appendix A.7.

Task	**Assets**	**Market Trend**	**Models**	**CR %↑**	**SR↑**	**MDD %↓**
Single Stock	GOOG	General Bullish ↗	w/ CVaR	25.077	1.052	17.530
			w/o CVaR	-1.461	-0.006	27.079
	NIO	General Bearish ↘	w/ CVaR	17.461	0.335	40.647
			w/o CVaR	-52.887	-1.002	70.243
Portfolio Management	(TSLA, MSFT, PFE)	Mixed	w/ CVaR	113.836	3.269	16.163
			w/o CVaR	14.699	1.142	17.511

Table 4: Key metrics FINCON with vs. without implementing **CVaR for within-episode risk control.** The performance of FINCON with the implementation of CVaR won a leading performance in both single-asset trading and portfolio management tasks.

Task	**Assets**	**Market Trend**	**Models**	**CR %↑**	**SR↑**	**MDD %↓**
Single Stock	GOOG	General Bullish ↗	w/ belief	25.077	1.052	17.530
			w/o belief	-11.944	-0.496	29.309
	NIO	General Bearish ↘	w/ belief	17.461	0.335	40.647
			w/o belief	8.197	0.156	55.688
Portfolio Management	(TSLA, MSFT, PFE)	Mixed	w/ belief	113.836	3.269	16.163
			w/o belief	28.432	1.181	27.535

Table 5: Key metrics FINCON with vs. without implementing **belief updates for over-episode risk control.** The performance of FINCON with the implementation of CVRF won a leading performance in both single-asset trading and portfolio management tasks.

5 Conclusion

In this paper, we present FINCON, a novel LLM-based multi-agent framework for financial decision-making tasks, including single stock trading and portfolio management. Central to FINCON is the Synthesized Manager-Analyst hierarchical communication structure and a dual-level risk control component. This communication method channels financial data from multiple sources to specialized analyst agents, who distill it into key investment insights. The manager agent then synthesizes these insights for decision-making. Our experimental evaluations demonstrate the efficacy of our risk control mechanism in mitigating investment risks and enhancing trading performance. Additionally, the streamlined communication structure reduces overhead. The dual-level risk control component introduces a novel approach to defining agent personas, enabling dynamic updates of risk and market beliefs within agent communication. A valuable future research direction would be to scale FINCON's framework to manage large-sized portfolios comprising tens of assets, while maintaining the impressive decision-making quality demonstrated with smaller portfolios. Given the LLM's input length constraint, a critical challenge lies in striking an optimal balance between information conciseness through agent distillation and potential performance deterioration when extending the current context window. Addressing this will be essential for ensuring quality-assured outcomes.

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